

Technical Specification — “Koxlear” Backend System

1. Project Overview

The “Koxlear” project is a backend system designed to handle speech-to-text (STT) analysis, task management, and API-based interactions for students, teachers, and administrators. The system allows users to upload audio assignments, receive evaluations, and view performance results. Through the admin/backoffice interface, content, users, and analytical processes can be fully managed.

2. Project Goals and Requirements

2.1 Goals

- Provide students with the ability to improve pronunciation through audio-based assignments.
- Enable teachers to evaluate answers both automatically and manually.
- Give administrators full control over users, content, and analytics.
- Ensure backend scalability using Docker, Celery, and asynchronous processing.

2.2 Functional Requirements

User management, task creation, audio uploads, analysis, evaluations, admin control, and reporting via REST API.

2.3 Non-Functional Requirements

API response time: 1–2s per request. Scalability, PEP8 compliance, security (JWT, HTTPS, encryption), and monitoring.

3. Database Structure (Tentative)

Table	Main Fields	Description
User	id, username, email, password_hash, role, created_at, updated_at	Stores user information
Task	id, title, description, audio_file, created_by, due_date	Teacher-created tasks
AudioSubmission	id, task_id, student_id, audio_file, stt_text, feedback, score	Students' audio and results
Lesson	id, name, description, level, created_at, updated_at	Lessons or modules
LessonAssignment	id, lesson_id, student_id, assigned_at, completed_at	Lesson assignments
Notification	id, user_id, type, message, is_read, created_at	Notification records

4. API Endpoints

Endpoints for authentication, task management, audio uploads, evaluations, and notifications.

5. Asynchronous Tasks

Celery handles STT analysis asynchronously with Redis as broker and cache.

6. Security and Permissions

JWT authentication, role-based access, HTTPS, encryption, and logging.

7. Technologies and Environment

Django REST, Celery, Redis, PostgreSQL, Docker, CI/CD, monitoring.

8. Project Stages

Requirements, design, implementation, async setup, testing, deployment, documentation.

9. Expected Outcomes

Stable, secure, scalable backend with complete API and STT integration.

10. Special Integrations

STT module, notifications, analytics dashboard, versioning, file cleanup.